

Decimals Worksheet 3 – 1st ESO

Topic: Competency Problems Related to FM Radio Frequencies

Level: 1st Year Secondary Education (1st ESO)

1. Theoretical Summary

Decimal numbers represent parts of a whole, expressed using a decimal point. For example, 87.5 MHz means 87 whole units and 5 tenths of a unit. When working with radio frequencies, decimals help us tune the radio more precisely.

2. Exercises

1. Order these frequencies from smallest to largest:
87.5 MHz, 89.3 MHz, 88.75 MHz, 90.1 MHz
 2. Round these frequencies to the nearest whole MHz:
 - 87.5 MHz
 - 89.3 MHz
 - 88.75 MHz
 - 90.1 MHz
 3. Convert the following frequencies from MHz to kHz (1 MHz = 1000 kHz):
 - 87.5 MHz
 - 89.3 MHz
 4. Convert these frequencies to Hz (1 MHz = 1,000,000 Hz):
 - 87.5 MHz
 - 90.1 MHz
 5. If your radio can tune up to 0.05 MHz (50 kHz) steps, how many steps are there between 87.5 MHz and 90.0 MHz?
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3. Competency Problems

1. You scanned a radio station at 88.75 MHz but saved it as 88.7 MHz. What is the error in MHz and in kHz?
2. If your radio antenna receives a signal at 89.35 MHz, explain how decimals help you identify the correct station compared to just 89 MHz.
3. Imagine your school project needs to save frequencies with three decimal places. Write an example frequency and explain how precise it is compared to only one decimal place.

4. You have saved 10 frequencies on your radio. List 3 ways decimals can help you organize or compare these saved stations effectively.
5. The radio module tunes frequencies between 76.0 MHz and 108.0 MHz. Calculate the total range in MHz and convert it to kHz.

Solutions

Exercises:

1. Ordered frequencies: 87.5 MHz, 88.75 MHz, 89.3 MHz, 90.1 MHz
2. Rounded:
 - 87.5 MHz → 88 MHz
 - 89.3 MHz → 89 MHz
 - 88.75 MHz → 89 MHz
 - 90.1 MHz → 90 MHz
3. MHz to kHz:
 - 87.5 MHz = 87,500 kHz
 - 89.3 MHz = 89,300 kHz
4. MHz to Hz:
 - 87.5 MHz = 87,500,000 Hz
 - 90.1 MHz = 90,100,000 Hz
5. Number of steps:
(90.0 - 87.5) MHz = 2.5 MHz
2.5 MHz / 0.05 MHz = 50 steps

Competency Problems:

1. Error = 88.75 - 88.7 = 0.05 MHz = 50 kHz
 2. Decimals allow tuning more precisely, distinguishing 89.35 MHz from 89.0 MHz, avoiding station confusion.
 3. Example: 88.756 MHz is precise to the thousandth, much more exact than 88.7 MHz which is only precise to the tenth.
 4. Use decimals to:
 - Order stations by frequency
 - Identify closest stations without overlap
 - Save precise frequencies for future tuning
 5. Range = 108.0 - 76.0 = 32.0 MHz = 32,000 kHz
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